Moed A 2021 solution

1 Standing waves

At $x = 0$ the string is tied to wall, and at $x = L$ tied to a massless ring on a frictionless pole.

1) Write the NORMALIZED basis functions

$$\psi(x,t) = \sin(\omega t + \phi)\left(A\cos(kx) + B\sin(kx)\right)$$

$$\psi(0,t) = 0$$

$$A = 0$$

$$\psi'(L,t) = 0$$

$$\cos(kL) = 0$$

$$k_n L = \frac{\pi}{2} + \pi n = \frac{\pi}{2}(2n+1)$$

$$\boxed{k_n = \frac{\pi}{2L}(2n+1)}$$

$$u_n = A\sin(k_n x)$$

$$A^2 \int_0^L \sin^2(k_n x)\, dx = 1$$

$$A^2 \frac{L}{2} = 1$$

$$A = \sqrt{\frac{2}{L}}$$

$$\boxed{u_n = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi}{2L}\left(2n+1\right)x\right)}$$

2) Initial conditions:

$$\psi\left(x,0\right) = a\sin\left(\frac{\pi}{2L}x\right) + b\sin\left(\frac{5\pi}{2L}x\right)$$

$$\dot{\psi}\left(x,0\right) = 0$$

$$\psi\left(x,t\right) = \sqrt{\frac{2}{L}} \sum_{n=0} B_n \sin\left(k_n x\right) \sin\left(ck_n t + \phi_n\right)$$

At $t = 0$

$$\psi\left(x,0\right) = \sum_{n=0} B_n \sin\left(k_n x\right) \sin\left(\phi_n\right)$$

$$a\sqrt{\frac{2}{L}} \sin\left(\frac{\pi}{2L}x\right) + b\sqrt{\frac{2}{L}} \sin\left(\frac{5\pi}{2L}x\right) = \sqrt{\frac{2}{L}} \sum_{n=0} B_n \sin\left(k_n x\right) \sin\left(\phi_n\right)$$

$$a\underbrace{\langle u_m | u_0 \rangle}_{\delta_{m,0}} + b\underbrace{\langle u_m | u_2 \rangle}_{\delta_{m,2}} = \sin\left(\phi_m\right) B_m$$

$$B_0 = \frac{a}{\sin\left(\phi_0\right)}$$

$$B_2 = \frac{b}{\sin\left(\phi_2\right)}$$

Initial condition for velocity:

$$0 = c \sum_{n=0} k_n \cos(\phi_n) B_n \sin(k_n x)$$

$$\boxed{\phi_n = \frac{\pi}{2}}$$

$$B_0 = a$$

$$B_2 = b$$

$$\boxed{\psi(x,t) = a \sin(k_0 x) \sin\left(ck_0 t + \frac{\pi}{2}\right) + b \sin(k_2 x) \sin\left(ck_2 t + \frac{\pi}{2}\right)}$$

3)

$$ck_0 t + \frac{\pi}{2} = \pi p$$

$$ck_0 t + \frac{\pi}{2} = \pi l$$

where l, p are integers.

$$t = \frac{L}{c}(2p - 1)$$

$$t = \frac{L}{5c}(2l - 1)$$

Both of the above equations must be satisfied:

$$\frac{L}{c}(2p - 1) = \frac{L}{5c}(2l - 1)$$

$$5\left(2p-1\right)=\left(2l-1\right)$$

$$10p-5=2l-1$$

$$10p=2l+4$$

$$l=5p-2$$

If $p=1$ then $l=3$

$$\boxed{\tau=\frac{L}{c}}$$

4)

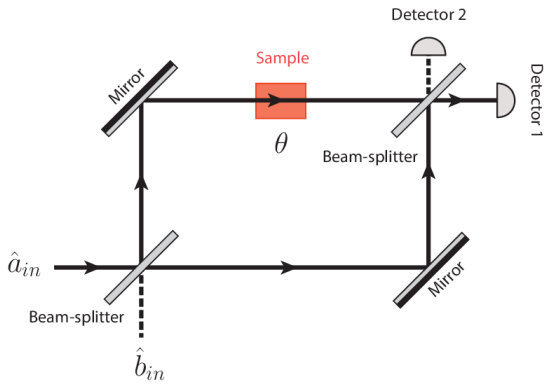
When $t=\tau$ the string is tied to the ground at $x=\frac{L}{2}$.

The new wave functions:

$$\boxed{u_n=\sqrt{\frac{4}{L}}\sin\left(\frac{\pi}{L}\left(2n+1\right)x\right)}$$

2 Mach-Zehnder interferometer

2.1



Wikipedia: The Mach–Zehnder interferometer is a device used to determine the relative phase shift variations between two collimated beams derived by splitting light from a single source.

2.2

The device is most sensitive in the close to where the derivative is largest (in absolute value).

For detector 1:

$$I_1\left(\phi_0+\Delta\phi\right) \approx I_1\big|_{\phi_0} + \left.\frac{dI_1}{d\phi}\right|_{\phi_0} \Delta\phi$$

$$\Delta I_1 \approx \left.\frac{dI_1}{d\phi}\right|_{\phi_0} \Delta\phi$$

We need to find where $\frac{dI_1}{d\phi}$ assumes its largest (absolute) value:

$$\frac{d^2 I_1}{d\phi^2} = 0$$

$$\frac{d}{d\phi}\sin\left(\phi\right)\cos\left(\phi\right) = 0$$

$$\frac{d}{d\phi}\sin(2\phi) = 0$$

$$\cos(2\phi) = 0$$

$$\boxed{\phi_0 = \pm\frac{\pi}{4}}$$

For detector 2:

$$\frac{d^2 I_2}{d\phi^2} = 0$$

$$\frac{d}{d\phi}\sin(\phi)\cos(\phi) = 0$$

$$\Downarrow$$

$$\boxed{\phi_0 = \pm\frac{\pi}{4}}$$

The device is most sensitive in the neighborhood of $\phi_0 = \pm\frac{\pi}{4}$.

2.3

The light that travels through the arm with the gas will accumulate a phase that depends on the index of refraction of the gas, which varies with pressure:

$$n(\omega; P) = \sqrt{1 + \frac{Ney_0}{\epsilon E_0}}$$

where N is volumetric electron density.

Ideal gas law:

$$PV = N_e k_B T$$

where N_e is the number of electrons. The density of electrons N is then:

$$N = \frac{N_e}{V} = \frac{P}{k_B T}$$

$$\Downarrow$$

$$n\left(\omega; P\right) = \sqrt{1 + \frac{ey_0}{\epsilon E_0}\frac{P}{k_B T}}$$

The phase ϕ:

$$\phi = L\Delta k = Lk_0\left(1 - \sqrt{1 + \frac{ey_0}{\epsilon E_0}\frac{P}{k_B T}}\right)$$

The (normalized) signal at detector 1:

$$I_1 = \cos^2\left(Lk_0\left(1 - \sqrt{1 + \frac{ey_0}{\epsilon E_0}\frac{P}{k_B T}}\right)\right)$$